

Lumateperax Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-362/C

Process Validation Summary

Process performance qualification was executed on three consecutive commercial-scale batches. All predefined acceptance criteria for critical process parameters and critical quality attributes were met, supporting a conclusion of a validated state of control.

Cleaning verification swab results were below the health-based exposure limits derived for the product. No changes to the approved analytical procedures are proposed in this submission. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable.

Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. The control strategy links critical quality attributes to critical process parameters identified during development. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable.

Manufacturing Process Overview

The drug product is manufactured by a conventional wet granulation process comprising dispensing, high-shear granulation, fluid-bed drying, milling, blending, compression, and film coating. In-process controls are applied at each unit operation, including loss on drying after fluid-bed drying and blend uniformity prior to compression. Process parameters were established during process performance qualification and are maintained under the site quality system.

Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. No changes to the approved analytical procedures are proposed in this submission. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection.

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Control of Excipients (3.2.P.4)

For the commercial formulation F-362/C, Solutab Type A (Croscarmellose sodium, disintegrant) is used.

Grade selection and control follow the principles of ICH Q8(R2) and Ph. Eur. general monograph 2034.

Formulation Development Rationale

During formulation development a number of croscarmellose sodium grades were evaluated. The development team ultimately proceeded with Solutab Type A; batches manufactured with this material consistently exhibited tighter particle size distribution supporting content uniformity, which the alternative grades evaluated did not achieve at comparable levels.

Analytical Method Validation Summary

The assay and related-substances procedures employ reversed-phase high performance liquid chromatography with ultraviolet detection. Validation demonstrated acceptable specificity, linearity, accuracy, precision, and robustness. Forced degradation studies confirmed the stability-indicating capability of the related-substances method.

Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. No changes to the approved analytical procedures are proposed in this submission. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control.

Regional Information

No excipients of human or animal origin are used in the drug product with the exception of lactose monohydrate, for which a TSE/BSE compliance declaration is provided in Module 3.2.A.2. Certificates of suitability are maintained where applicable.

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Container Closure System

The commercial packaging configuration consists of high-density polyethylene bottles with child-resistant polypropylene closures and induction seal liners. Container closure integrity was demonstrated by dye ingress testing. The packaging components comply with applicable food-contact regulations and compendial requirements for plastic packaging systems.

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Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Cleaning verification swab results were below the health-based exposure limits derived for the product. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. No changes to the approved analytical procedures are proposed in this submission.

Stability Summary

Primary stability studies were conducted under ICH long-term (25°C/60% RH) and accelerated (40°C/75% RH) conditions on three registration batches. All quality attributes remained within acceptance criteria through the tested intervals. Photostability testing per ICH Q1B confirmed that the product is not photolabile in the commercial pack.

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Reference Standards

Primary reference standards for the drug substance and specified impurities are qualified against compendial standards where available. Working standards are qualified against the primary standard under an approved protocol and are stored under controlled conditions.

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Current Commercial Specification

Component	Function	Grade per current specification
Croscarmellose sodium	disintegrant	Ac-Di-Sol SD-711
Reference		SPEC-EX-5540 Rev 02

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Control of Drug Product

The drug product specification includes tests for description, identification, assay, uniformity of dosage units, dissolution, related substances, water content, and microbial limits. Analytical procedures are compendial where available; non-compendial procedures were validated in accordance with ICH Q2(R1).

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